

Problem A

Harmonic progression sum

The simplest harmonic progression is

$$1/1, 1/2, 1/3, 1/4, 1/5, \dots$$

Let $S_n = \sum_{i=1}^n (1/i)$, compute this sum to arbitrary precision after the decimal point.

Input

The input contains only one test case. The first line contains two values: the first is the number of digits D and the second is the value of N . Consider $(1 \leq D \leq 10^5)$ and $(1 \leq N \leq 10^8)$.

The input must be read from the standard input.

Output

The output contains only one line printing the value of the sum with exact D precision.

The output must be written to the standard output.

Example

Input example 1	Output example 1
12 7	2.592857142857